MIS772 – Predictive Analytics

T2 2024

Assignment 2 – Individual
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Executive summary (1 page)

Executive problem statement:

Airbnb is facing two key business challenges that impact guest satisfaction and overall property performance. First, there is uncertainty around whether the way hosts describe their properties has any real influence on how guests perceive and rate their stay. Understanding this relationship is crucial in ensuring that hosts are setting the right expectations, which can ultimately impact guest reviews and future bookings.

Second, Airbnb needs to predict future property review scores to identify which aspects of a property or host service are most closely tied to guest satisfaction. This would allow hosts to make targeted improvements to their properties and services, thereby enhancing guest experiences and increasing positive reviews.

The objective is to use the available data to analyse these two issues—determine if there is a relationship between host descriptions and guest reviews and develop a reliable model to predict future review scores. Solving these problems will help Airbnb improve property listings and guide hosts to optimize guest satisfaction.

Executive solution statement:

In response to Airbnb's inquiry, the analysis for Task A showed that there is a very **weak correlation** (**see Figure 1** for detailed correlation analysis) between the sentiment scores derived from host property descriptions and the sentiment scores from customer reviews. This indicates that while host descriptions might influence guest expectations, they are not a primary factor in shaping how guests ultimately rate their experiences. Based on this, it is recommended that Airbnb could introduce standardized guidelines or a checklist for hosts when writing property descriptions. This would ensure consistency in the way key aspects of the property. Another recommendation is that Airbnb could introduce a feedback mechanism where hosts receive regular insights on how their descriptions compare to actual guest experiences. This ongoing feedback loop could help hosts continually refine their descriptions to better align with guest expectations and experiences.

For Task B, a predictive model was developed using key attributes such as property features and different types of ratings to estimate future review scores. The model demonstrated strong reliability during validation and can be used by Airbnb to provide actionable insights to hosts on how to improve their properties to enhance guest satisfaction. Specifically, properties with higher cleanliness and better communication tend to score higher, and focusing on these areas will likely result in better overall guest reviews.

The anticipated benefits of these solutions for Airbnb include improved property descriptions that better manage guest expectations, leading to more positive reviews. Additionally, the predictive model will enable Airbnb to guide hosts toward targeted improvements, ultimately increasing guest satisfaction and retention, thereby driving higher bookings and platform engagement.

Data exploration, pattern discovery, and preparation (2 pages)

Task A

The task was to find the correlation between the sentiment scores of both host and customer. The dataset was derived from two excel files “**listings_va.xlsx**” and **reviews_va.xlsx**. bridged by “**Property ID**”

Data exploration and preparation

In preparation for this analysis, careful attention was given to ensuring that only relevant and complete data was used to achieve accurate results. Initially, both the property descriptions and customer reviews were extracted from the dataset using the “**Select Attributes**” operator, which streamlined the dataset to only the necessary fields. To maintain data integrity, entries with missing descriptions were systematically removed using the “**Filter Examples**” operator, enhancing the overall consistency and accuracy of the data.

Since the selected attributes, specifically the descriptions and reviews, were originally in nominal form, the "**Nominal to Numerical**" operator was employed to transform these into a text format for quantitative analysis. Additionally, the "**Set Role**" operator was used to designate the property ID as the unique identifier. This step was essential in ensuring proper linkage between datasets in subsequent operations, allowing the model to accurately connect individual properties with their corresponding descriptions and reviews. (**See Figure 2**)

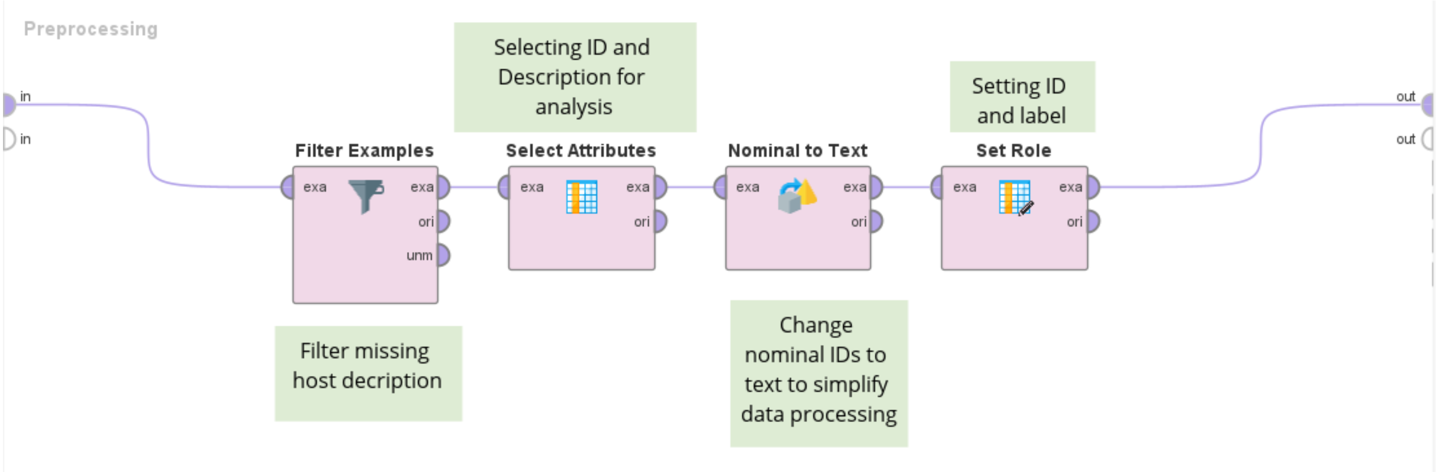

Figure 2

Text processing

The sentiment analysis was performed through the "**Process from Documen**t" operator, which involved a structured series of steps. Initially, a predefined sentiment text file was imported, categorizing words as either positive or negative. This approach ensured uniformity in sentiment classification, allowing for consistent and accurate sentiment scoring across all property descriptions and customer reviews in the dataset. Process included, (see **figure 3**)

- **Transform Cases:** The text is first converted to lowercase. This ensures that all text is standardized and that variations in capitalization do not affect further analysis, such as tokenization or word matching.
- **Tokenize:** This step splits the text into individual tokens, typically words. Tokenization breaks the continuous stream of text into manageable components that can be analysed individually. Each token can then be analysed or transformed in subsequent steps.
- **Stem (Snowball):** In this step, words are reduced to their root or base form. This process, known as stemming, allows for grouping words with the same base meaning. This reduces redundancy and focuses the analysis on the core meaning of words, simplifying the sentiment calculation.
- **Filter Stop words**: This step removes common stop words, which are words that generally do not contribute meaning to the text (e.g., "and," "the," "is"). These words are removed to reduce noise in the data and focus the analysis on words that contribute meaning and sentiment.
- **Filter Tokens (by Length):** Tokens (words) that are too short or too long are removed in this step. Very short tokens (like single letters) often provide little value in analysis, while excessively long tokens may not be meaningful. By filtering based on length, the analysis focuses on tokens of an appropriate length that are more likely to provide meaningful insight.

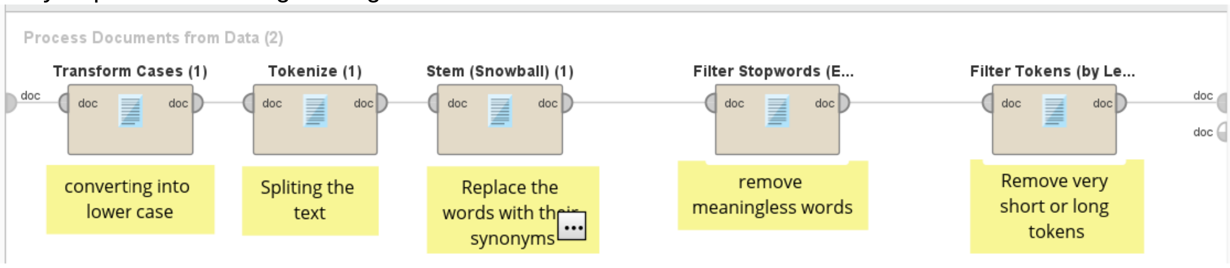

Figure 3

Sentiment Analysis

After this, “**Generate Aggregation**” was used to count the count of both negative and positive words from both the description and customer reviews. The “**Generate attribute**” operator was used to calculate the sentiment score (count of positive words – count of negative words). “**Join**” operator then connected the newly generated scores both the host and reviews linking them with the property ID. This joined dataset was then used for the generating the correlation matrix. (**See figure 5**)

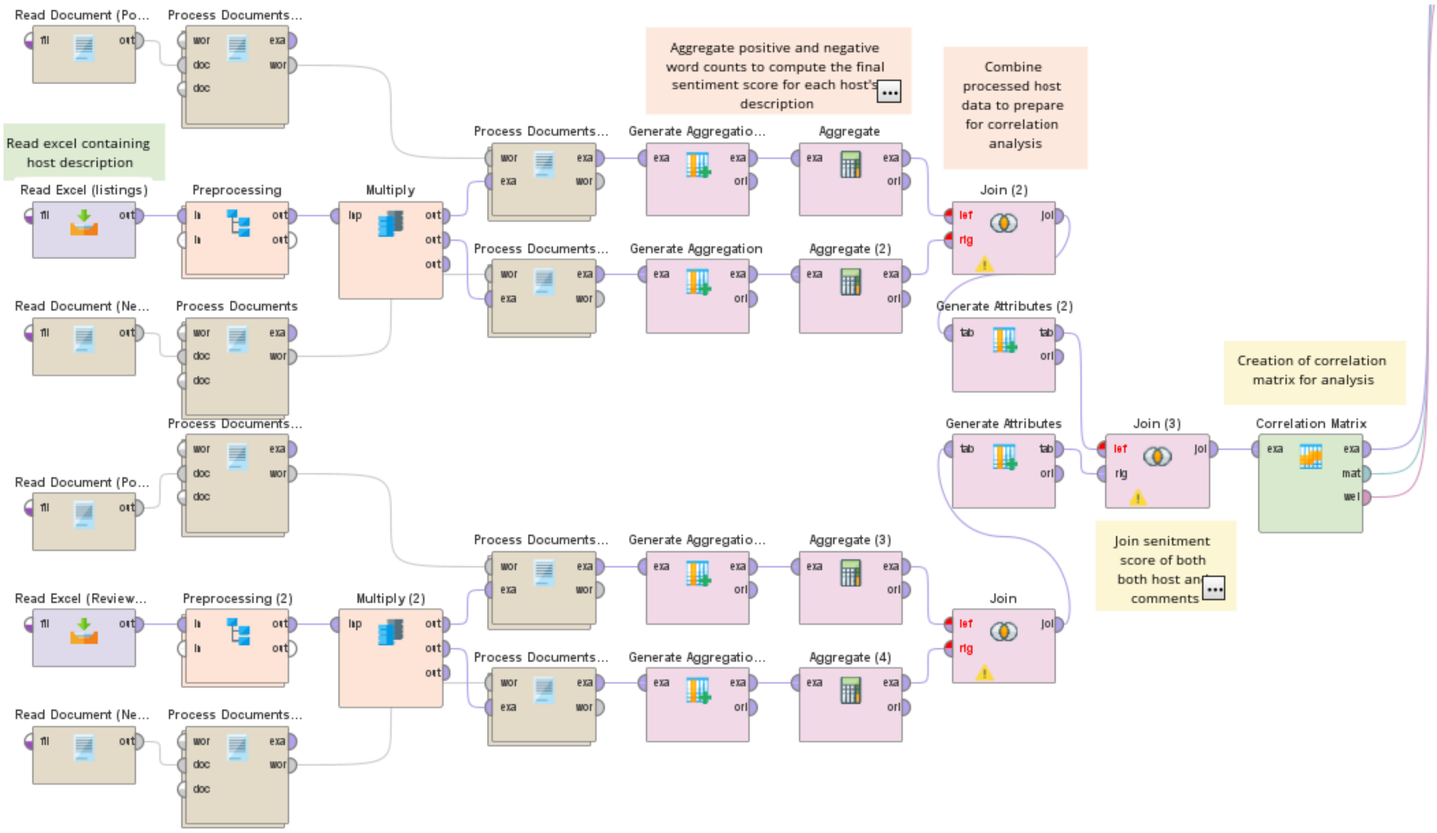

Figure 5

Task B

Data exploration

The task involved developing a predictive model to estimate the "**Review_score_Rating**" based on various attributes from the dataset "**listings_va.xlsx**," which included a mix of numeric, nominal, and text-based attributes. The target variable for prediction is the **review_scores_rating**, with selected predictor variables encompassing property details, host characteristics, and customer data. The model-building process was meticulously designed to satisfy the foundational assumptions of linear regression, including ensuring linear relationships, minimizing multicollinearity, and achieving normal distribution of the data with homoscedasticity (equal variance). Anomaly detection techniques were also employed to identify and manage outliers, preserving the model's robustness and accuracy.

Data preprocessing

The predictive model employs linear regression, necessitating that all data be transformed into a numerical format. To identify the relationship between various predictors and the **review_scores_rating,** a correlation matrix (**see figure 6**) was first generated, highlighting the most influential predictors (see Figure 6). The "**Select Attributes**" operator was then applied to isolate the most relevant features for the model. While **reviews_per_month** exhibited a strong correlation, it was excluded due to a **p-value (>0.05)** exceeding the significance threshold, indicating insufficient statistical relevance. Additionally, certain attributes with minimal correlation were retained in the model based on their potential strategic business importance.

Attributes	id	host_tot...	bedroo...	review_...	review_...	review_...	review_...	review_...	review_...	revie...	instant_...
id	1	0.062	0.018	-0.082	-0.090	-0.062	-0.098	-0.099	-0.056	-0.081	0.191
host_total_listings_count	0.062	1	0.024	-0.109	-0.087	-0.045	-0.097	-0.099	-0.019	-0.088	0.065
bedrooms	0.018	0.024	1	0.015	0.002	0.025	-0.012	-0.003	0.013	0.006	0.030
review_scores_rating	-0.082	-0.109	0.015	1	0.785	0.737	0.653	0.696	0.474	0.814	-0.093
review_scores_accuracy	-0.090	-0.087	0.002	0.785	1	0.676	0.644	0.671	0.495	0.734	-0.073
review_scores_cleanliness	-0.062	-0.045	0.025	0.737	0.676	1	0.532	0.552	0.419	0.695	-0.046
review_scores_checkin	-0.098	-0.097	-0.012	0.653	0.644	0.532	1	0.735	0.475	0.587	-0.076
review_scores_communication	-0.099	-0.099	-0.003	0.696	0.671	0.552	0.735	1	0.477	0.613	-0.074
review_scores_location	-0.056	-0.019	0.013	0.474	0.495	0.419	0.475	0.477	1	0.476	0.020
review_scores_value	-0.081	-0.088	0.006	0.814	0.734	0.695	0.587	0.613	0.476	1	-0.081
instant_bookable	0.191	0.065	0.030	-0.093	-0.073	-0.046	-0.076	-0.074	0.020	-0.081	1

figure 6

Handling Missing Values.

To ensure model accuracy, the target variable**, review_scores_rating,** was meticulously filtered to remove any missing values using the "**Filter Examples**" operator. Additionally, **host_total_listings_count** was also filtered due to the minimal impact of its missing values. For the remaining variables, missing data was replaced with the median, a method chosen to enhance resilience against outliers, thereby improving the robustness and reliability of the predictive model. These preprocessing steps were crucial to maintaining data completeness and preventing potential bias or inaccuracies, ensuring the integrity of the subsequent analysis.

Nominal to numerical conversion

In the model preparation phase, nominal values, such as **'instant_bookable'**, were transformed into numeric formats to facilitate analysis by the linear regression model. This conversion utilized the 'unique integers' method, where each distinct nominal value is systematically assigned a consecutive number. This approach enables the linear regression model to interpret categorical data effectively, enhancing its ability to generate precise predictions.

Normalising Attributes

The data was then normalised used “**Normalise**” operator so that the dataset is standardised and follow the assumptions that the dataset has a normal distribution which is the core idea of linear regression. And also, when it comes to anomaly detection, the data points would be equally weighted while outliers are being detected.

Anomaly detection

In order to ensure equal variance, the outlier in the dataset needs to be removed from the dataset. To find the outliers The **k-NN Global Anomaly Detection** operator was used. This operator is very efficient against the visual inspection. The operator was used with a k-value of 5, which implies that each data point’s distance from its five nearest neighbors was calculated. This means any point with a distance score of 5 will be identified as outlier. (**see Figure 7**)

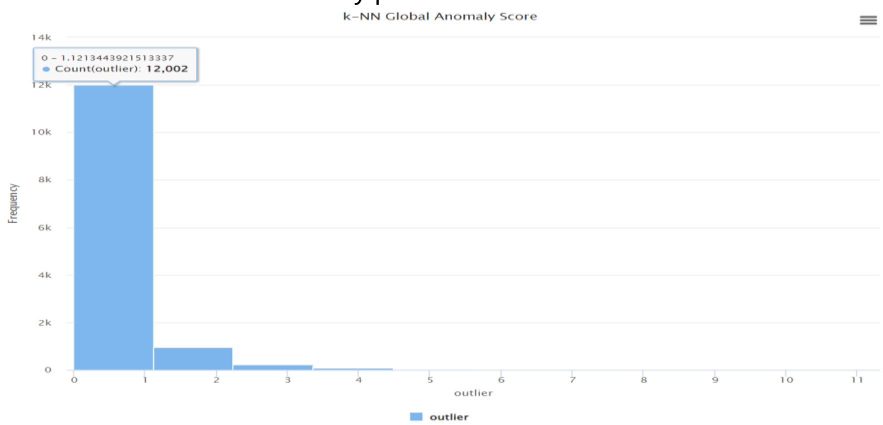

Figure 7

The **figure 8** illustrates the data processing steps

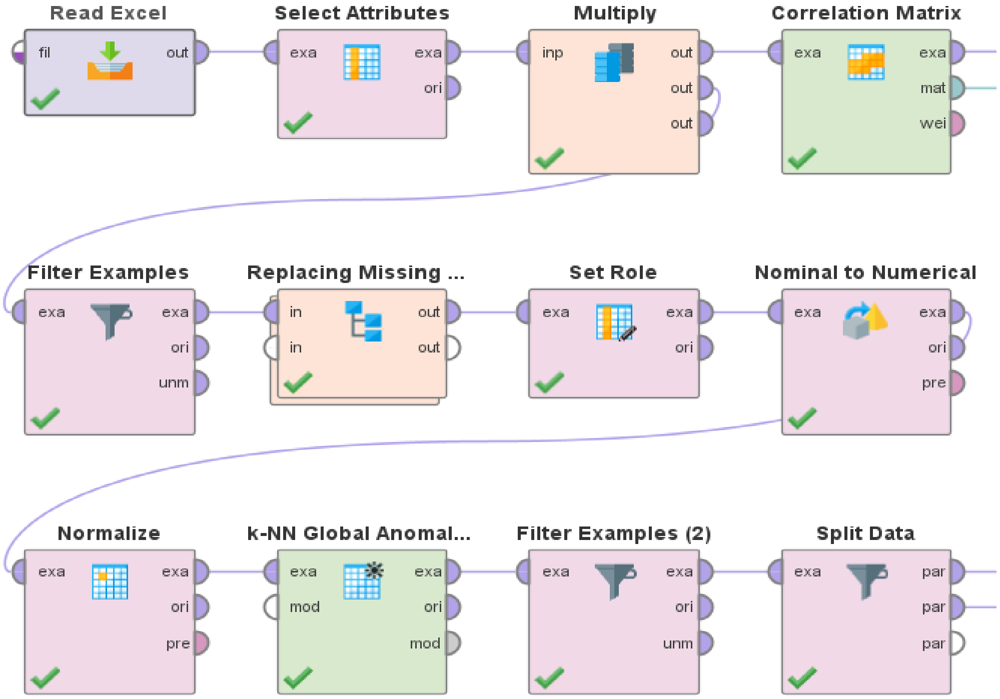

figure 8

Predictive modelling (2 pages)

For the modelling, the pre-processed data is used and splitted into two parts with a ratio of 7:3. This means 70% of the data will be used for training and the rest 30% will be used for testing. The "Split" operator is used to split the data. Hence, ensuring that the model can be used against new data, giving a realistic prediction. The linear regression model was selected due its effectiveness of prediction against numerical data. The model predicted the target variable using predictors selected. The result (see **figure 9**) showed the relationship of selected variables and the target variable.

LinearRegression

```
- 0.242 * instant_bookable
+ 3.564 * review_scores_value
+ 0.115 * bedrooms
+ 2.316 * review_scores_accuracy
+ 2.327 * review_scores_cleanliness
+ 0.611 * review_scores_checkin
+ 1.490 * review_scores_communication
- 0.212 * host_total_listings_count
+ 92.448
```

figure 9

The training and testing data was applied the linear regression using the "**Apply model**" operator and the performance was evaluated using the "**Performance** (**regression**)" operator. The metrics used to evaluate the performance was **Root Mean Squared Error (RMSE)** and **Root Relative Squared Error (RRSE)**. The RMSE recorded 4.4533 and RRSE 0.501. This means that the deviations of the model predictions from the actual data is 4.533 on an average and the relative error rate is **50.1%** (refer **figure 10**)

PerformanceVector

```
PerformanceVector:
root_mean_squared_error: 4.533 +/- 0.000
root_relative_squared_error: 0.501
```

figure 10

Model Parameters

The model can be justified because the p- value of all the selected attributes falls below the significance threshold **(<0.05**) which ensures the statistical significance of the model. (Refer **figure 11**)

Attribute	Coefficient	Std. Error	Std. Coefficient	Tolerance	t-Stat	p-Value
instant_bookable	-0.242	0.049	-0.025	0.990	-4.897	0.000
review_scores_value	3.564	0.080	0.353	0.382	44.404	0
bedrooms	0.115	0.048	0.012	1.000	2.366	0.018
review_scores_accur...	2.316	0.087	0.227	0.342	26.738	0
review_scores_cleanli...	2.327	0.074	0.233	0.448	31.571	0
review_scores_check...	0.611	0.083	0.059	0.485	7.379	0.000
review_scores_com...	1.490	0.086	0.142	0.474	17.427	0
host_total_listings_cou...	-0.212	0.053	-0.020	0.992	-3.999	0.000
(Intercept)	92.448	0.048	?	?	1939.112	0

figure 11

Model evaluation and improvement (2 pages)

After creating the model for predicting the rating, it was evaluated using different validation methods. The **Hold-out validation** and **cross validation** with fold value of 5 and 10. The results showed variation in performance which was recorded and helped in selecting the final model.

Model Comparisons

The models created was compared using the performance metrics ie, **RMSE and RRSE**. The Hold-out validation **RMSE** was **4.533** and **RRSE 0.501**(refer **figure 10**). This is the best results compared against cross validation. The cross validation with fold value of **10** showed **RMSE** of **4.566** and **RRSE** of **0.482.** (refer **figure 12**). While cross validation of fold value of **5** yeilded **RMSE** of **4.574** and **RRSE** of **0.481** (refer **figure 13**)

PerformanceVector

```
PerformanceVector:
root_mean_squared_error: 4.566 +/- 0.348 (micro average: 4.578 +/- 0.000)
root_relative_squared_error: 0.482 +/- 0.037 (micro average: 0.481)
```

Figure 12

PerformanceVector

```
PerformanceVector:
root_mean_squared_error: 4.574 +/- 0.191 (micro average: 4.577 +/- 0.000)
root_relative_squared_error: 0.481 +/- 0.017 (micro average: 0.480)
```

figure 13

Improvement Measures

To improve the model, various validation method was used. The performance metrics between all the validation methods was very close to each other yet the hold out validation method was better in terms of performance. Additionally, the attributed were modified such as review_per_month attribute showed a high correlation between the target variable yet the p value in the model showed that it was not statistically significant. Hence it was removed which enhanced the model prediction. Another improvement measure was identifying the outliers in the dataset and removing them which might misdirect the model due to extreme values.

The linear regression model indeed showed a good prediction model with both the validation methods and can be easily recommended to Airbnb to predict future ratings. This model will help them with decision making process and better guest satisfaction.